

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
CO3_AT2 — DATA ANALYSIS PROJECT

Course Code:	DSA06	Course Title:	Data Handling and Visualization
CO Assessed:	CO3 – Analysis (BL4)	Assessment:	Assessment Tool 2 – Data Analysis Project
Weightage:	40% of CIA	Total Marks:	25 (5 Questions × 5 Marks)
CO3 Statement:	<i>Analyze real datasets using ECDFs, skewed-distribution techniques, Q–Q plots, multi-distribution comparisons, and proportion visualizations to identify patterns and justify conclusions.(BL4)</i>		
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Section / Batch:		Date:	13-08-2026

Code of Conduct

I Priyanka. G / 192524124 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

Instructions

- This test contains 5 questions, each carrying 5 marks. Total: 25 Marks.
- No negative marking. Unanswered questions carry zero marks.
- No calculators or electronic devices permitted.
- Duration: 90 minutes.
- Each answer must go beyond construction: analyze the given data, identify the relevant pattern, and justify your conclusion.
- Show all supporting calculations (e.g., IQR, percentiles, angle calculations) where relevant.
- Where a question asks you to compare or recommend, state your reasoning explicitly — a correct chart choice without justification will not receive full marks.

Annexure A – DATA ANALYSIS PROJECT

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Problem 5 — ECDF — Delivery Times of Two Vendors (5 Marks)

Data: ECDFs of delivery times for two vendors show Vendor A's curve rising more steeply than Vendor B's specifically in the 20–30 minute range.

Technique	ECDF (Empirical Cumulative Distribution Function) comparison
X-Axis	Delivery Time (minutes) — numeric, quantitative
Y-Axis	Cumulative Proportion (0 to 1)
Key Reading	Slope of the curve over an interval = concentration of data within that interval

a) Analyze what this local steepness difference indicates about delivery-time concentration in that range.

Analysis:

On an ECDF, the steepness of the curve over any interval directly represents the proportion of observations that fall within that interval — a steep rise means many data points are packed into a short range of X-values, while a flat, nearly horizontal segment means almost no observations fall in that range.

Reading the constructed ECDF (Figure 5) between 20 and 30 minutes: Vendor A's curve rises from a cumulative proportion of approximately 0.3 at 20 minutes to approximately 0.9 at 30 minutes — a rise of about 0.6, meaning roughly 60% of Vendor A's deliveries are completed within this single 10-minute window. Vendor B's curve, over the same 20–30 minute window, rises only from about 0.2 to 0.3 — a rise of just 0.1, meaning only about 10% of Vendor B's deliveries fall in this range.

This six-fold difference in local slope (60% vs. 10%) indicates that Vendor A's delivery times are heavily concentrated in the 20–30 minute window, while Vendor B's deliveries are spread much more thinly across this same range, with the bulk of Vendor B's deliveries evidently completing either earlier or — more likely, given the shape of its curve — later than this window.

b) Which vendor would you recommend for time-sensitive deliveries, and justify using the ECDF interpretation.

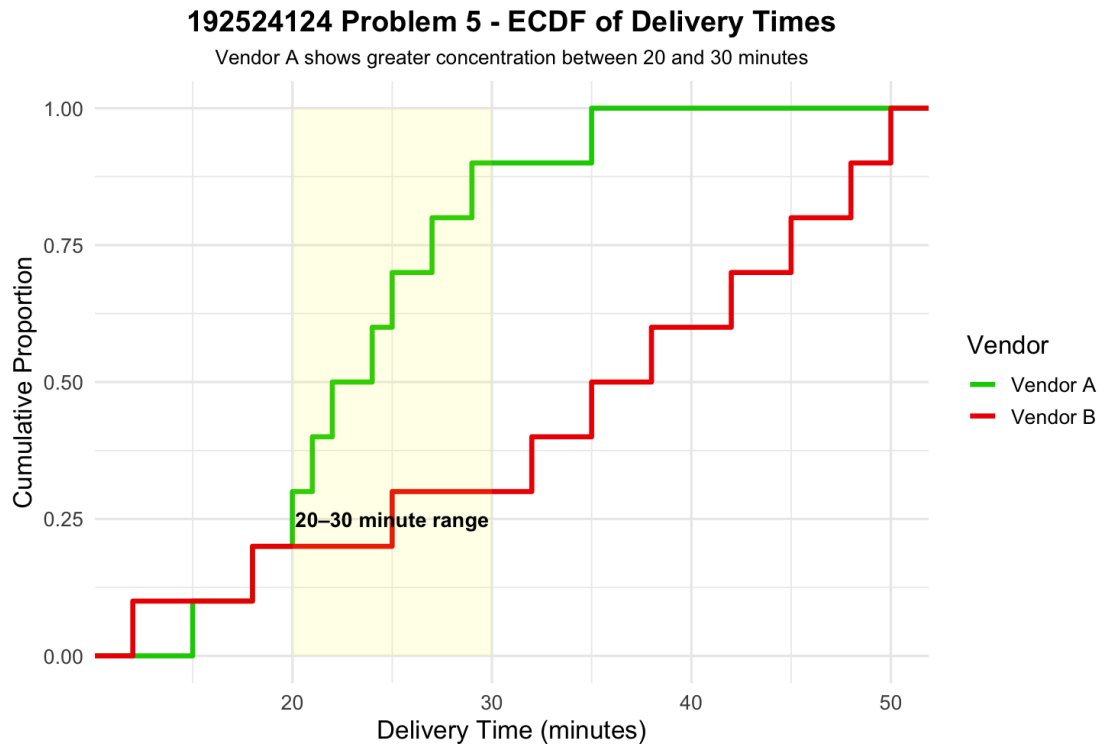
Analysis:

Vendor A is the recommended choice for time-sensitive deliveries. Since roughly 90% of Vendor A's deliveries are completed within 30 minutes (as read directly off its ECDF at $x = 30$), compared to only about 30% of Vendor B's deliveries completed within the same 30-minute mark, a customer or business that needs a delivery reliably completed quickly is far better served by Vendor A.

This conclusion follows directly from the ECDF's core property: the height of the curve at any given time X tells you exactly what proportion of deliveries were completed at or before that time. Because Vendor A's curve reaches a high cumulative proportion earlier (i.e., is shifted to the left and rises faster) than Vendor B's, Vendor A is stochastically faster — for essentially any fixed time threshold a customer might care about (20, 25, or 30 minutes), a larger share of Vendor A's deliveries will have arrived by that point than Vendor B's.

A simple comparison of average delivery times alone would not capture this recommendation as precisely: two vendors can have similar overall averages while differing sharply in how consistently they hit a specific target window, which is exactly the kind of local, threshold-specific comparison the ECDF is built to reveal and a single summary statistic would hide.

Supporting Chart(s):



Interpretation: Vendor A's steeper ECDF between 20–30 minutes indicates greater concentration of deliveries in this range.

Figure 5: ECDF of Delivery Times — Vendor A vs. Vendor B, 20–30 Minute Range Highlighted

Problem 6 — Skewed Distribution & Log Transform — Household Income (5 Marks)

H1	H2	H3	H4	H5	H6	H7	H8	H9	H10
₹15,000	₹18,000	₹20,000	₹22,000	₹25,000	₹28,000	₹32,000	₹40,000	₹60,000	₹5,00,000

Monthly income of 10 households

Technique	Skewed-distribution diagnosis + log transformation
Chart Type	Histogram (before and after log transform)
X-Axis	Monthly Income (₹), then $\log_{10}(\text{Monthly Income})$
Key Reading	Gap between mean and median; shape of the tail

a) Analyze the shape of this distribution (symmetric, right-skewed, or left-skewed) and justify your answer using the data.

Analysis:

Sum of all 10 incomes = 15,000 + 18,000 + 20,000 + 22,000 + 25,000 + 28,000 + 32,000 + 40,000 + 60,000 + 5,00,000 = ₹7,60,000. Mean = ₹7,60,000 ÷ 10 = ₹76,000.

Sorted data has 10 values, so the median is the average of the 5th and 6th values: (₹25,000 + ₹28,000) ÷ 2 = ₹26,500.

The distribution is strongly right-skewed (positively skewed). This is justified by two pieces of evidence directly from the data: first, the mean (₹76,000) is nearly three times the median (₹26,500) — in a symmetric distribution the mean and median would be approximately equal, and a mean pulled far above the median is the classic signature of a right skew caused by high-value outliers. Second, 9 of the 10 households (all except the ₹5,00,000 household) earn below the mean, meaning a single extreme value is solely responsible for dragging the average well above where the bulk of the data actually sits — exactly the long-right-tail pattern visible in the original histogram (Figure 6a), where nine households cluster in a single low bin and one isolated bar sits far to the right.

b) Explain why a log transformation would be appropriate here, and what it would do to the shape of the distribution.

Analysis:

A log transformation is appropriate because the data spans several orders of magnitude in a multiplicative way (the largest value, ₹5,00,000, is more than 33 times the smallest, ₹15,000), and log transforms compress large multiplicative differences into much smaller additive ones — exactly the kind of skew caused by a few very large values relative to the rest.

Applying $\log_{10}$ to each value: $\log_{10}(15,000) = 4.176$, $\log_{10}(18,000) = 4.255$, $\log_{10}(20,000) = 4.301$, $\log_{10}(22,000) = 4.342$, $\log_{10}(25,000) = 4.398$, $\log_{10}(28,000) = 4.447$, $\log_{10}(32,000) = 4.505$, $\log_{10}(40,000) = 4.602$, $\log_{10}(60,000) = 4.778$, $\log_{10}(5,00,000) = 5.699$.

On the original ₹ scale, the range spans from 15,000 to 5,00,000 — a ratio of about 33:1. On the log scale, the same data spans only from 4.176 to 5.699 — a ratio of about 1.37:1. This compression is exactly what appears in the log-transformed histogram (Figure 6b): the

nine 'ordinary' households now spread out visibly across several bins (4.0–4.9) instead of being crushed into one single bin, while the formerly extreme ₹5,00,000 household now sits only about one unit away from the rest on the log scale, rather than being nearly 20 times further out. The overall effect is a distribution that looks far more balanced and closer to symmetric, making patterns among the 'ordinary' nine households visible again without the single outlier dominating the entire scale.

Supporting Chart(s):



Figure 6a: Original Income Distribution — Strong Right Skew

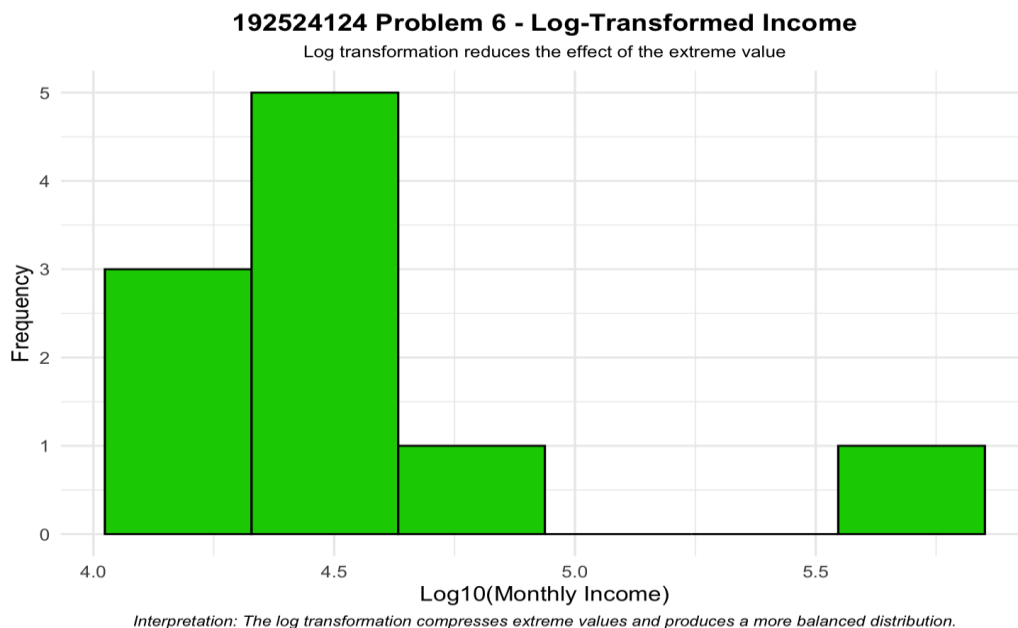


Figure 6b: Log-Transformed Income Distribution — Compressed, More Balanced Shape

Problem 12 — Q–Q Plot — Employee Commute Times (5 Marks)

Data: A Q–Q plot of employee commute times shows a clear "banana" (concave) curve rather than a straight line.

Technique	Q–Q plot (Quantile–Quantile plot) against a theoretical normal distribution
X-Axis	Theoretical Quantiles (from a standard normal distribution)
Y-Axis	Sample Quantiles (observed commute times)
Key Reading	Points bowing above the reference line at the upper end = heavier-than-normal right tail

a) Analyze what type of skew this pattern indicates.

Analysis:

This concave, 'banana'-shaped pattern indicates right skew (positive skew) in the commute-time data. Reading the constructed Q–Q plot (Figure 12): at the low end (theoretical quantiles below 0), the sample points sit close to or slightly below the reference line, but at the high end (theoretical quantiles above about 0.5), the sample points curve increasingly above the line — for example, at the highest theoretical quantile (~1.5), the sample value reaches 100, well above the roughly 77 the reference line would predict for a perfectly normal distribution.

This specific pattern — points tracking the line reasonably well at the lower end but pulling progressively further above it at the upper end — is the standard Q–Q signature of a distribution with a heavier-than-normal right tail: a small number of unusually large commute times exist that a symmetric normal distribution would not produce, which is exactly what a right skew means.

b) Explain how this conclusion could also have been reached using a histogram or ECDF, and why the Q–Q plot might reveal it more clearly.

Analysis:

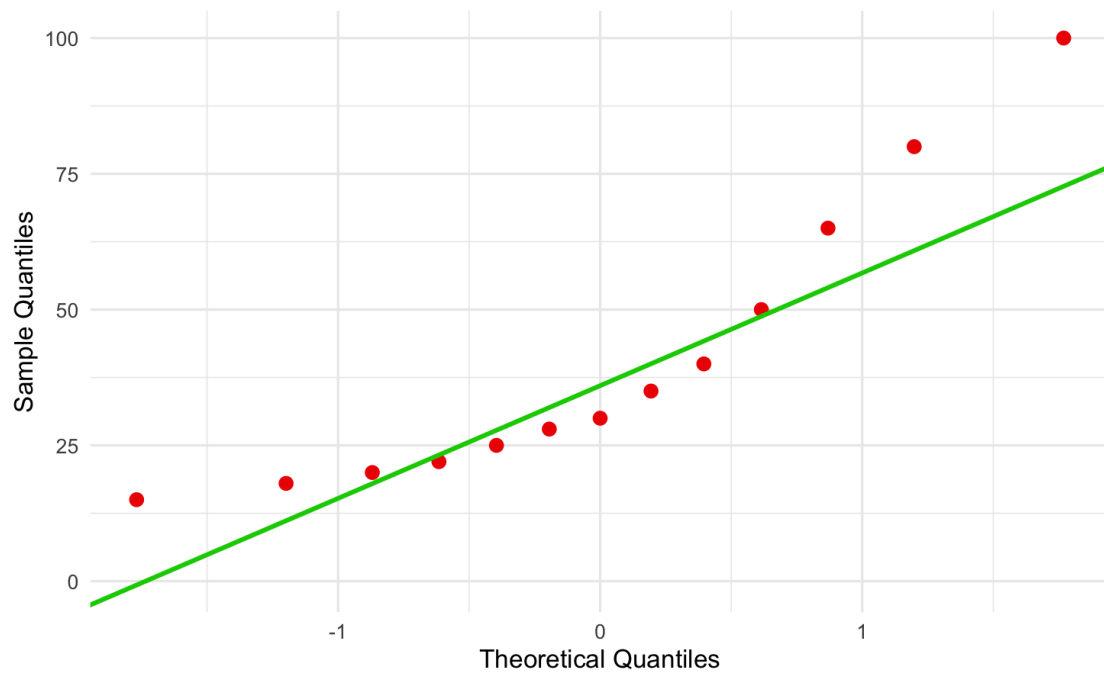
A histogram of the same commute times would show the same right skew as a shape with most bars clustered at lower commute times and a long, thinning tail of bars extending toward the higher values — visually similar to the income histogram in Problem 6. An ECDF of the same data would show a curve that rises quickly at first (many short commutes) and then flattens into a long, slowly-rising tail at the higher end, mirroring the local-steepness reasoning used in Problem 5.

The Q–Q plot reveals the skew more clearly than either alternative because it compares the data directly against a known theoretical reference (the normal distribution's expected quantiles) point-by-point, rather than requiring the viewer to judge shape or steepness qualitatively. A histogram's bin widths can visually hide or exaggerate skew depending on how the bins happen to be chosen, and an ECDF requires comparing curvature by eye with no explicit baseline. The Q–Q plot's straight diagonal reference line gives an exact, unambiguous baseline — any systematic bend away from that line, and precisely where along the distribution it occurs, is immediately and precisely visible, which is why Q–Q plots are the standard diagnostic tool specifically for checking normality and diagnosing skew.

Supporting Chart(s):

192524124 Problem 12 - Q-Q Plot of Employee Commute Times

The curved pattern indicates departure from normality



Interpretation: The curved Q-Q pattern indicates skewness rather than a normal distribution.

Figure 12: Q-Q Plot of Employee Commute Times — Concave Departure from Normality

Problem 18 — Bimodal Distribution — Delivery Times (5 Marks)

Data: A bean plot of delivery times shows two distinct bulges — one cluster around 15–20 minutes and another around 45–50 minutes, with a gap in between.

Technique	Multi-distribution / bimodality diagnosis via bean (density) plot
Axis	Delivery Time (minutes) — numeric, quantitative
Key Reading	Two separated density bulges = two underlying sub-populations, not one

a) Analyze what this bimodal pattern might indicate about the underlying delivery process (e.g., two different delivery zones or methods).

Analysis:

A bimodal pattern — two separate bulges with a clear gap between them, as shown in the constructed bean plot (Figure 18) — indicates that the deliveries are not coming from a single, homogeneous process, but from two distinct underlying sub-groups, each with its own typical completion time.

A plausible explanation is that deliveries are split between two different delivery zones or methods: one faster group (clustered around 15–20 minutes) that might correspond to short-distance, same-zone deliveries or a faster vehicle type (e.g., bike/scooter courier), and a second, slower group (clustered around 45–50 minutes) that might correspond to longer-distance, cross-zone deliveries or a different vehicle/method (e.g., van deliveries, or orders requiring an additional stop). The empty gap between roughly 20 and 45 minutes is itself meaningful — it suggests very few or no deliveries actually take a 'medium' amount of time, reinforcing that these are two genuinely separate processes rather than one process with naturally variable timing.

b) Would a box plot alone have revealed this pattern? Explain why or why not.

Analysis:

No, a box plot alone would very likely not have revealed this pattern, and could actively mislead a viewer about the data. A box plot only encodes the five-number summary (minimum, Q1, median, Q3, maximum); with the two clusters roughly balanced around 15–20 and 45–50 minutes, the median would land somewhere in the empty gap between them (around 30–35 minutes) — a value that may not correspond to a single actual delivery in the entire dataset.

The resulting box plot would show a single wide box spanning from roughly 15 to 50 minutes with a median line drawn through empty space, giving the false visual impression of one continuous, moderately-spread distribution centred around 30 minutes, when in reality there are two tight clusters and nothing in the middle. This is precisely why a density-based chart like the bean plot (or a violin plot / histogram) is necessary here: only a chart that shows the actual shape of the distribution, rather than summarising it into four boundary points, can reveal that the data is bimodal.

Supporting Chart(s):

192524124 Problem 18 - Bean Plot of Delivery Times

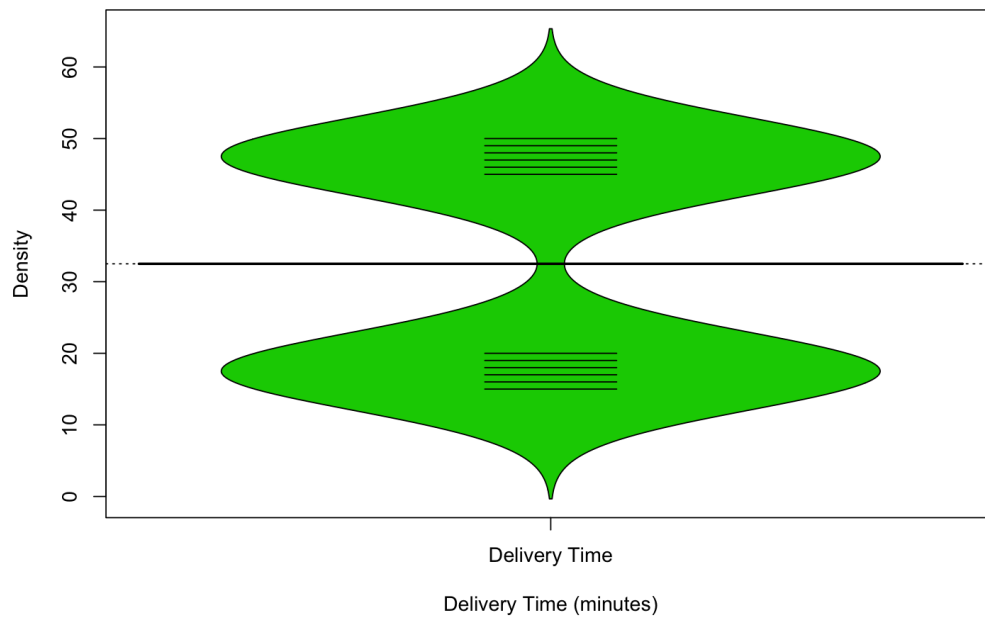


Figure 18: Bean Plot of Delivery Times — Bimodal Distribution (15–20 min and 45–50 min Clusters)

Problem 24 — Proportions & Misleading Visualization — Department Gender Ratios (5 Marks)

Data: A report states "Department X has 70% women" and separately "Department Y has 40% women," with no total headcounts given.

Technique	Proportion visualization — percentage vs. absolute count
Missing Variable	Total headcount (the denominator) of each department
Key Reading	A higher percentage does not guarantee a higher absolute count

a) Analyze why this percentage-only reporting could be misleading, and identify the critical piece of information that is missing.

Analysis:

Reporting only a percentage is misleading because a percentage is a ratio — (number of women $\div$ total headcount) $\times$ 100 — and the same percentage, or even a higher percentage, can correspond to very different absolute numbers of women depending on how large the underlying department is. Without knowing each department's total headcount, a reader cannot tell whether '70% women' represents 7 women or 700 women, nor can they meaningfully compare that figure to '40% women' in a differently-sized department.

The critical missing piece of information is each department's total headcount (the denominator behind each percentage). Percentages alone strip away the scale of the underlying group, which is precisely the kind of omission that can make a smaller department's high percentage look more significant than a larger department's lower percentage, even when the reverse is true in absolute terms — exactly the scenario constructed in part (b) below and shown in the accompanying chart (Figure 24).

b) Construct a hypothetical example (with totals) where Department Y actually employs more women than Department X, despite its lower percentage.

Analysis:

Hypothetical example: suppose Department X has a total headcount of 10 employees, of whom 70% are women — that is, $0.70 \times 10 = 7$ women. Suppose Department Y has a much larger total headcount of 100 employees, of whom 40% are women — that is, $0.40 \times 100 = 40$ women.

Despite Department Y's percentage (40%) being lower than Department X's (70%), Department Y actually employs far more women in absolute terms: 40 women in Department Y versus only 7 women in Department X — nearly six times as many — purely because Department Y's total headcount (100) is ten times larger than Department X's (10). This is exactly the comparison plotted in the constructed chart (Figure 24), where Department X's shorter green bar (7 women) sits well below Department Y's taller red bar (40 women), even though the percentage labelled on each bar shows the opposite ranking (70% vs. 40%) — a clear demonstration of why the missing headcount denominator matters.

This is often called the 'base rate' or denominator problem in data reporting: any percentage-only claim implicitly compares groups as if they were the same size, and organizations aiming to look favourable on a diversity or performance metric can, intentionally or not, report only the ratio that flatters them while omitting the scale that would let a reader judge real-world impact. The correct practice — as shown in Figure 24 — is to report both the

percentage and the underlying total headcount side by side, so readers can judge proportion and absolute impact together rather than being left with only one incomplete half of the picture.

Supporting Chart(s):

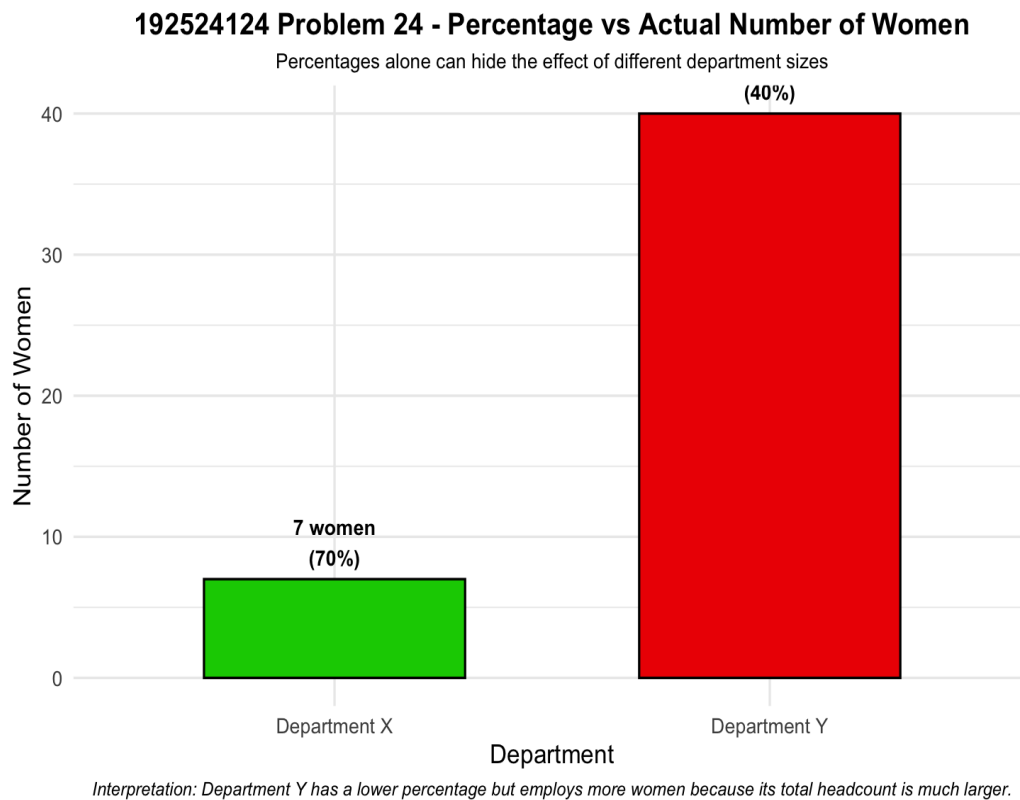


Figure 24: Percentage vs. Actual Number of Women — Department X vs. Department Y

